Task:
Finding the majority bit of a 5-bit binary string like 10101, 10010, 00011...

Idea:
The bit which will appear at least 3 times, is the majority bit out of 5 bits. So, we can find if 1 appeared 3 times or not. If yes, 1 will be majority and vice versa.

Naive approach:
We know, AND gate is true when all the inputs are true. That is, we will get 1 as output, only if all three inputs are 1. So, we can consider all the three bit combinations from the binary strings.
Lets say, the inputs are ABCDE. So, combinations are:
F(A,B,C,D,E) = ABC + ABD + ABE + ACD + ACE +ADE
+ BCD + BCE + BDE
+ CDE
If any term of this function is true, we can say, 1 is the majority. Else, 0 is.
But, it will require us 9 OR gates and 20 AND gates in total. Which is a lot.

Optimized approach:
We know, AND gate outputs 1 when both input is 1 and XOR gate outputs 1 when there is odd number of 1 in input. That is,
A . B = 1, when A = 1 and B = 1
A $\oplus$ B = 1, when either A = 1 or B = 1.

Lets say inputs are ABCDE. We make three groups AB, CD and E. Here, we see that, there are three possible cases of getting at least three 1s (for being majority).

Case – 1:
A and B both 1. And, any of the rest of the inputs is 1.

A = 1
B = 1
C + D + E = 1

Total = three 1s

Case – 2:
C and D both 1. And, any of the rest of the inputs is 1.

C = 1
D = 1
A + B + E = 1

Total = three 1s

Case – 3:

Either A or B is 1.
And, C or D is 1.
E is 1.

Total = three 1s

Finally,

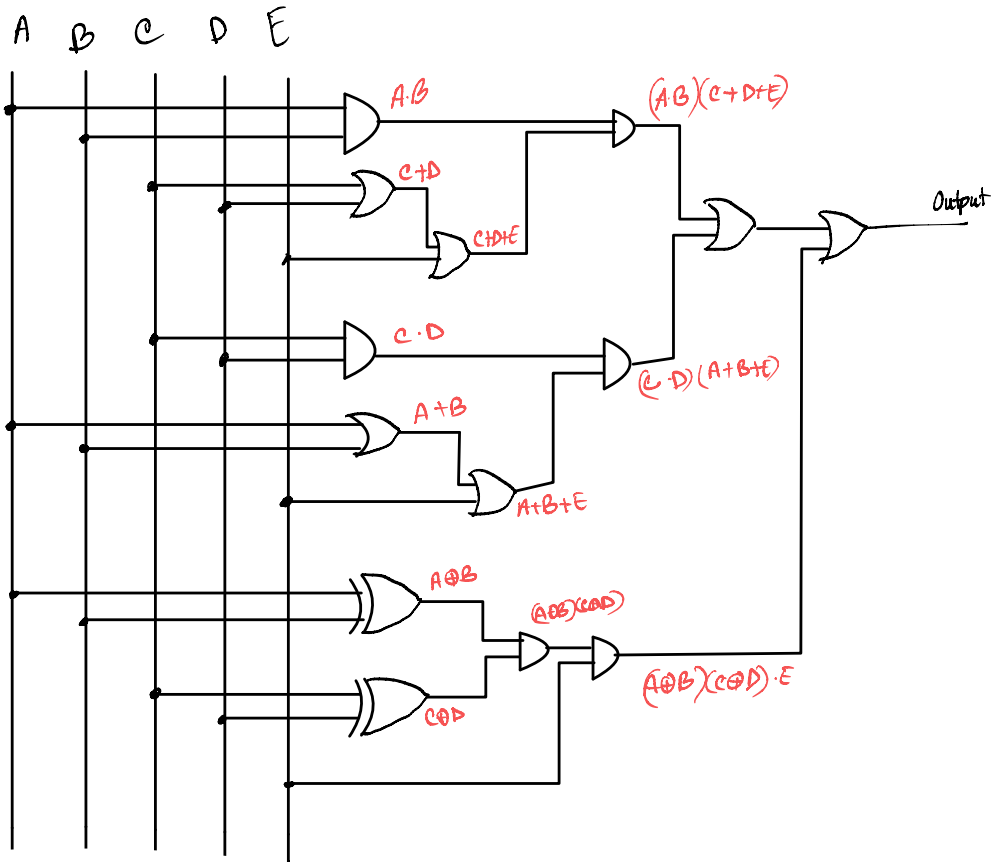
we can write these three cases in one equation,

$$F(A, B, C, D, E) = (A.B) . (C + D + E) + (C.D) . (A + B + E) + (A \oplus B) . (C \oplus D) . E$$

If any of the term is true, we will get 1 as the output and majority bit will be 1. Using this approach, we will need only 6 AND gates, 6 OR gates and 2 XOR gates.

Circuit Diagram:

F(A, B, C, D, E) = (A.B) . (C + D + E) + (C.D) . (A + B + E) + (A ⊕ B) . (C ⊕ D) . E



Final validity check:

Input (ABCDE)	Output / Majority Bit
00000	0
00001	0
00010	0
00011	0
00100	0
00101	0
00110	0
00111	1
01000	0
01001	0
01010	0
01011	1
01100	0
01101	1
01110	1
01111	1
10000	0
10001	0
10010	0
10011	1
10100	0
10101	1
10110	1
10111	1
11000	0
11001	1
11010	1
11011	1
11100	1
11101	1
11110	1
11111	1